

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a)     | <p><b>I</b> Temperature for all 7 volumes recorded.</p> <p><b>II</b> All thermometer readings including initial temperature (if recorded) ending in .0 or .5</p> <p><b>III</b> Selects two additional volumes of <b>FB 5</b> for experiments 6 and 7<br/> <b>AND</b> total volume for each additional experiment = 10.00 cm<sup>3</sup><br/> <b>AND</b> all four volumes recorded to 2 d.p. and ending in 0 or 5</p> <p><b>IV</b> Calculate the difference between the initial temperature of <b>FB 4</b> and the temperature recorded in experiment 3 for both the supervisor and the candidate.</p> <p>Mark awarded if difference between candidate temperature rise in experiment 3 and the supervisor temperature rise is less than or equal to 1.0 °C.</p> | <b>4</b> |
| 2(b)(i)  | <p><b>I</b> Unambiguous labelled axes with (maximum) temperature on the y-axis and volume of <b>FB 5</b> on the x-axis <b>OR</b> correct units<br/> <b>AND</b> suitable linear scales.</p> <p><b>II</b> Points plotted accurately.</p> <p><b>III</b> two lines of best fit drawn (straight line or smooth curve) that account for all the plotted points ignoring any points labelled anomalous</p> <p><b>IV</b> Both lines extrapolated to intersect.</p>                                                                                                                                                                                                                                                                                                      | <b>4</b> |
| 2(b)(ii) | Volume at intersection read correctly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>1</b> |
| 2(c)(i)  | <p>Correctly calculates<br/> amount <math>\text{HCl} = \frac{2.00 \times \text{volume from (b)(ii)}}{1000} \text{ mol}</math><br/> <b>AND</b> amount <math>\text{NaOH} = \text{amount HCl}</math><br/> <b>AND</b> both answers to 2 – 4 sf</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>1</b> |